

Letting

$$K = \frac{-g_{m1}R_1g_{m2}R_2}{(1+j\omega R_1C_1)(1+j\omega R_2C_2)} \quad (24.17)$$

we can rewrite Eq. (24.16) as

$$v_{out} = \frac{j\omega \frac{C_c}{g_{m1}}}{1+j\omega \frac{C_c}{g_{mcg}}} \cdot K \cdot v_{out} + K \cdot v_s \quad (24.18)$$

and, assuming $|K \cdot \frac{C_c}{g_{m1}}| \gg \frac{C_c}{g_{mcg}}$, thus

$$\frac{v_{out}}{v_s} \approx \frac{K \cdot \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{1 - j\omega \frac{C_c K}{g_{m1}}} \quad (24.19)$$

Resubstituting in K gives

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{(1+j\omega R_1C_1)(1+j\omega R_2C_2) + j\omega \frac{C_c}{g_{m1}} \cdot g_{m1}R_1g_{m2}R_2} \quad (24.20)$$

or, with $s = j\omega$, we can write

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + s \frac{C_c}{g_{mcg}}\right)}{s^2 \cdot (R_1C_1R_2C_2) + s \cdot (R_1C_1 + R_2C_2 + R_1g_{m2}R_2C_c) + 1} \quad (24.21)$$

Notice there is a LHP zero at

$$f_z = \frac{g_{mcg}}{2\pi C_c} \quad (24.22)$$

Since this zero is in the LHP, it will add to the phase response and enhance the speed of the op-amp. Intuitively, we can think that at high speeds the phase shift through C_c will cause the output signal to feed back and add to the signal at node 1. This *positive feedback* enhances the speed of the op-amp. To determine the location of the second pole, let's assume $R_1g_{m2}R_2C_c \gg R_1C_1$ or R_2C_2 or $R_1C_1R_2C_2$ so

$$s_{1,2} \approx \frac{-R_1g_{m2}R_2C_c \pm R_1g_{m2}R_2C_c}{2(R_1C_1R_2C_2)} \quad (24.23)$$

Our approximation won't tell us the location of the lower frequency pole (it's given by Eq. (21.65)). The location of the second pole is

$$f_2 \approx \frac{g_{m2}C_c}{2\pi \cdot C_1C_2} \approx \frac{g_{m2}C_c}{2\pi \cdot C_1(C_L + C_c)} \quad (24.24)$$

This result should be compared to Eq. (21.66). The location of the second pole is at a considerably higher frequency using this technique. *The result is that we can set the unity gain frequency to a higher value and still have a stable op-amp.* Further, the load capacitance (which is included in C_2) can be considerably larger for a given PM or GM. Again, Eq. (21.72) can be used to set the unity-gain frequency, f_{un} , assuming $f_2 \approx f_z$

$$f_{un} = \frac{g_{m1}}{2\pi C_c} (\approx f_z \text{ if } g_{m1} \approx g_{mcg}) \quad (24.25)$$

Using the *indirect compensation* method seen in Fig. 24.17 in the op-amp of Fig. 24.8 with $C_c = 240 \text{ fF}$ ($f_{un} = 100 \text{ MHz}$) gives the results seen in Fig. 24.19. The small-signal step response is seen in Fig. 24.20. This response should be compared to Figs. (24.12) and (24.14).

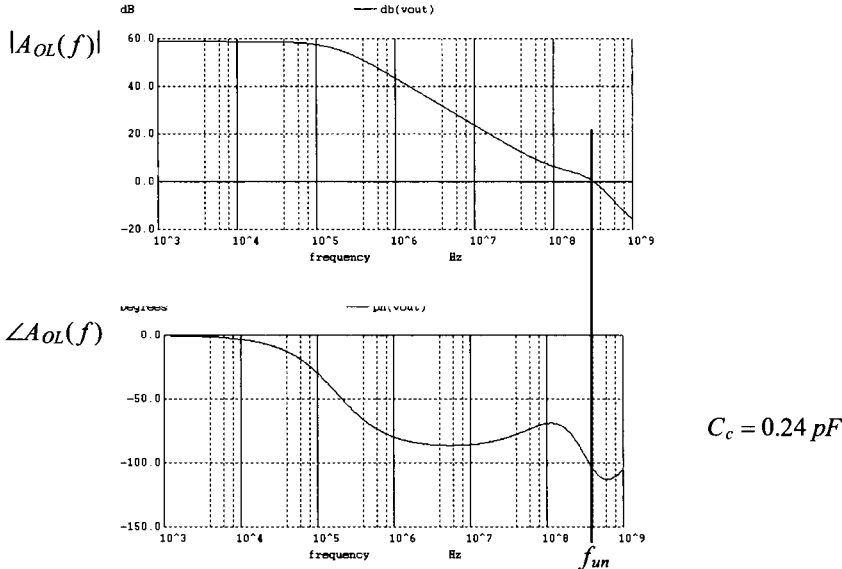


Figure 24.19 Simulating the op-amp of Fig. 24.8 using the indirect compensation scheme seen in Fig. 24.17 with a compensation capacitor of 240 fF.

The indirect feedback of the current through the compensation capacitor results in faster op-amp circuits and less layout area (the compensation capacitor generally dominates the layout area of an op-amp). However, the added circuit in Fig. 24.17 dissipates more power. To eliminate the additional power dissipation, consider the op-amp topology seen in Fig. 24.21. Here we've used the fact that a 100/2 PMOS device can be laid out as two 100/1 PMOS in series (see also Fig. 20.35). The current through the compensation capacitor is fed back through M4B to the output of the diff-amp. Note how we've also split M7 into two devices. It's important, for small offsets, to try to match both the drain-source and the gate-source voltages of the MOSFETs used in the op-amp. The step-response of the op-amp is seen in Fig. 24.22.

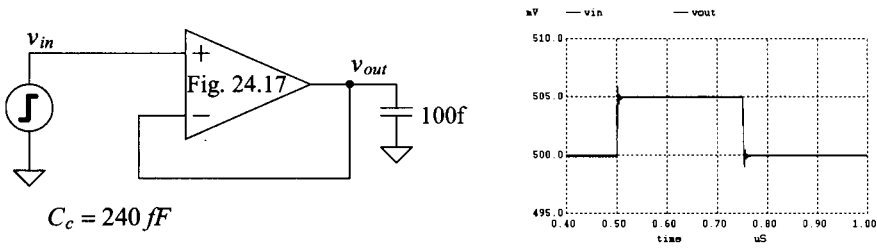


Figure 24.20 The step response of the op-amp in Fig. 24.17 with frequency response seen in Fig. 24.19.

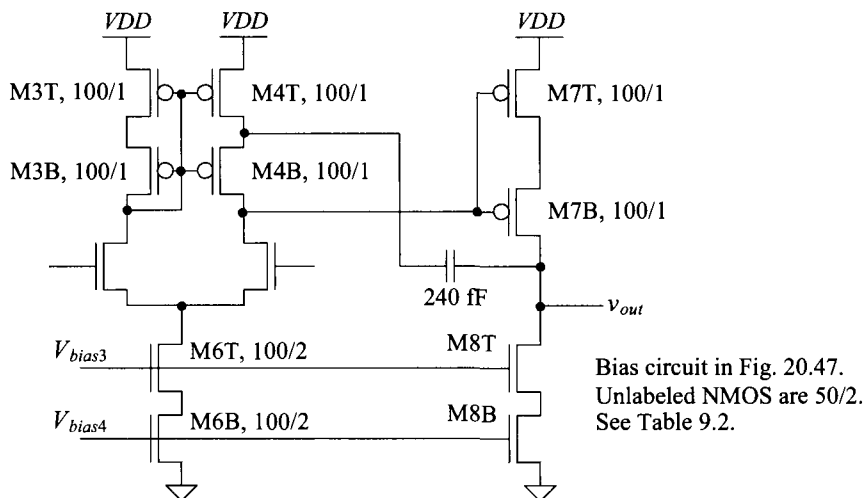


Figure 24.21 Implementing indirect feedback compensation without additional power dissipation in a two-stage op-amp.

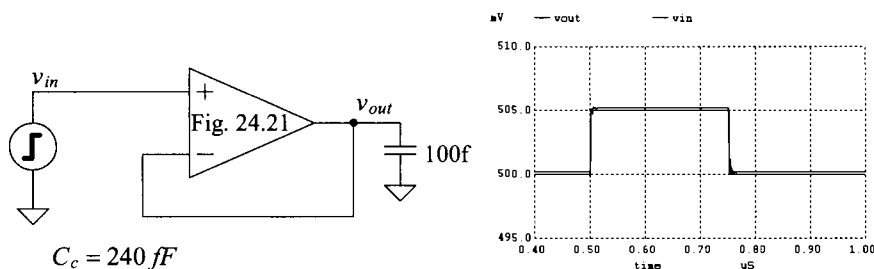


Figure 24.22 The step response of the op-amp in Fig. 24.21 driving 100 fF.

Slew-Rate Limitations

When we talked about step response, we limited the amplitude of the signals to small steps (5 mV) to avoid slew-rate limitations. Let's take the op-amp from Fig. 24.8 and put it in the configuration seen in Fig. 24.14, driving a 1 pF load capacitance. Further, let's increase the input pulse amplitude to (close to) the maximum allowable range, that is, from 500 mV to 900 mV (limited by the op-amp's input common-mode range). The simulated results are seen in Fig. 24.23. Referring to Fig. 24.8, notice that when the op-amp's noninverting input terminal (v_p) is driven high, M2 turns on and pulls the gate of M7 down. Since there isn't a current source in series with M7, it can quickly charge the 1 pF load capacitance. The only large-signal limitation is the bias current of the diff-amp, I_{SS} ($= 20 \mu\text{A}$ here), charging C_c ($= 2.4 \text{ pF}$ here). This limits the output rate of change since any variation in v_{out} causes a displacement current through C_c , which must be sourced/sunk by the diff-amp. This slew-rate limitation is calculated as $20 \mu\text{A}/2.4 \text{ pF}$ or 8.3 mV/ns. The change in the output voltage is 400 mV, so we would require roughly 50 ns to change the voltage across C_c .

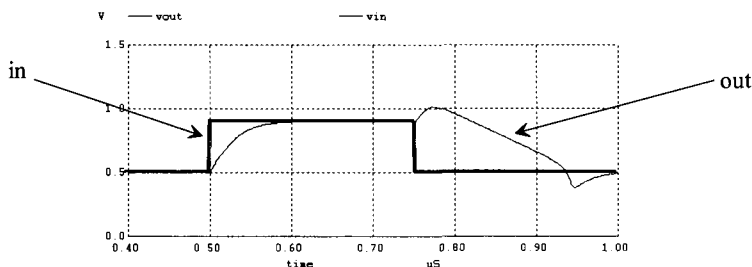


Figure 24.23 The large-signal (500 mV to 900 mV) performance of the op-amp in Fig. 24.8 with $C_c = 2.4$ pF and $R_z = 6.5$ k driving a 1 pF load. The low-to-high settling time is roughly 100 ns, while the high-to-low settling time, which is slew-rate limited, is roughly 500 ns.

The more interesting behavior occurs when the input signal drops from 900 mV to 500 mV. This causes M2 to shut off, allowing all of the current from M4 to charge the gate of M7. The result is a positive pulse on the gate of M7 that feeds directly to the output through the compensation capacitor (this is bad, and it is another example of why it is desirable to use the indirect compensation method). This pulse accounts for the (unwanted) positive movement in the output voltage seen in Fig. 24.23 just after the input pulse transitions negative. Now the load capacitor and the compensation capacitor must both be discharged through the constant current source M8. The output slew-rate is estimated by how fast this constant current (here 10 μ A) can discharge 3.4 pF (the sum of the compensation capacitor and the load capacitor). This rate is calculated as (roughly) 3 mV/ns. For the output to transition 400 mV requires 133 ns (this time will be longer because of the positive movement in the output voltage just after the input switches). Note that we might think that simply by sizing up the current conducting in M8 we can enhance the op-amp's slew-rate. This is true to the point where the diff-amp's current charging the compensation capacitor becomes the limiting factor.

Finally, note how the output voltage shoots past the final (desired) voltage level of 500 mV. This is because the gate of M7 is pulled to V_{DD} to shut it off when slewing is taking place (both M4 and M7 are shut off because of this). When the inputs of the op-amp move to the same value (500 mV here), the gate of M7 must be pulled downwards to the quiescent value. The time it takes for this to happen results in the undershoot in the op-amp's output voltage seen in Fig. 24.23.

The same test setup used to generate the data in Fig. 24.23 is also used to generate the data in Fig. 24.24 except that here the op-amp in Fig. 24.21 is used. Note the compressed time scale. The settling time for the low-to-high transition is now 10 ns, while the high-to-low transition time is 60 ns. We would expect much faster settling because the compensation capacitor is now ten times smaller than the value used in the op-amp of Fig. 24.8. Again, the settling time is limited by how fast M8 can discharge the load capacitance and the compensation capacitance. This can be estimated as $10 \mu\text{A}/1.24 \text{ pF}$ or 8 mV/ns. It takes 50 ns for the output to transition 400 mV. The size of M8 can be increased to meet a specific load capacitance and settling time requirement.

Because of the improved speed performance, smaller layout area, and (as we'll see) better power supply noise rejection, we'll use indirect compensation in place of Miller compensation in the remaining two-stage op-amps that we discuss in this book.

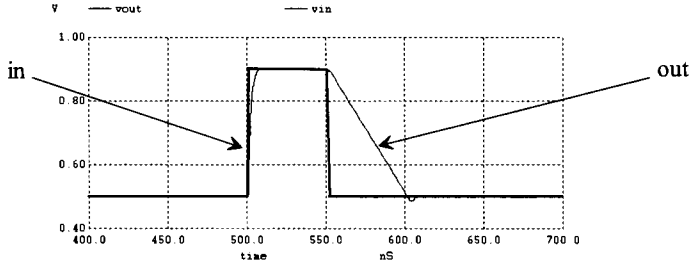


Figure 24.24 The large-signal (500 mV to 900 mV) performance of the op-amp in Fig. 24.21 with $C_c = 0.24$ pF driving a 1 pF load. The low-to-high settling time is roughly 10 ns, while the high-to-low settling time, which is slew-rate limited, is roughly 60 ns. Note the different time scale when compared to Fig. 24.23.

Common-Mode Rejection Ratio (CMRR)

The *CMRR* of an op-amp is calculated in the same way as the diff-amp in Sec. 22.1.3. The common-mode gain of the diff-amp is A_c . The common-mode gain of the op-amp is $A_c A_2$. The differential gain of the op-amp is $A_{OL}(f) = A_d \cdot A_2$ (where $A_d = A_1$). The *CMRR* of an op-amp in dB is given by

$$CMRR = 20 \cdot \log \left| \frac{A_{OL}(f)}{A_c \cdot A_2} \right| = 20 \cdot \log \left| \frac{A_d}{A_c} \right| \quad (24.26)$$

which shows that the op-amp *CMRR* is determined by the differential stage. Simulating the *CMRR* of an op-amp can be accomplished with the circuits seen in Fig. 24.25. For the op-amps discussed so far in this chapter, the *CMRR* is approximately 50 dB (see Ex. 22.7, where the common-mode voltage is 500 mV). We can think of A_c as the open-loop gain of the op-amp for common-mode signals. If the applied differential voltage is zero, and we change the common-mode voltage by ΔV_c , then the output voltage will change by ΔV_o .

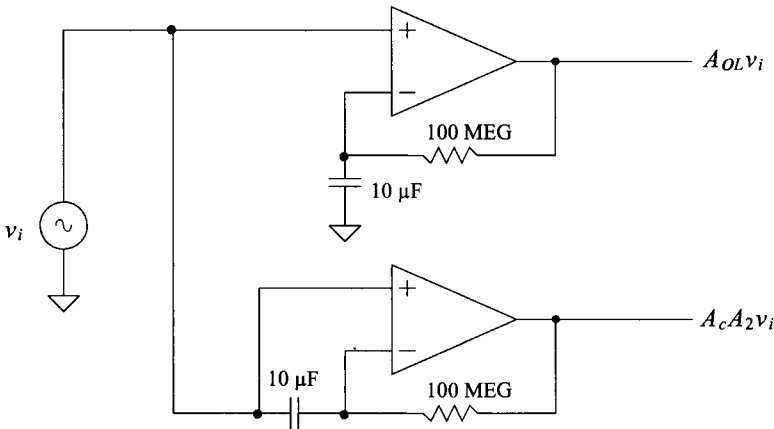


Figure 24.25 Circuit configuration used to simulate *CMRR*.

$= A_{cm} \cdot \Delta V_c$ (see Fig. 22.17). To compensate for the change in the output voltage, a nonzero input differential voltage develops on the input of the op-amp (an offset voltage that is a function of the common-mode voltage). This offset voltage can be estimated by

$$\Delta V_{OS} = \frac{\Delta V_o}{A_{OL}} = \frac{\Delta V_c \cdot A_{cm}}{A_{OL}} = \frac{\Delta V_c}{CMRR} \quad (24.27)$$

Knowing that 50 dB = 316, a change in the common-mode level on the inputs of the op-amp by 500 mV results in an input-referred offset voltage change of 500 mV/316 or 1.6 mV (if the gain of the diff-amp is 10, then its output voltage will change by 16 mV). This may not seem like such a big deal. However, notice that at higher frequencies, in Fig. 22.16, the $CMRR$ falls off, indicating that the common-mode gain is increasing. This leads to distortion and forces the use of op-amp topologies in which common-mode voltage doesn't vary (the inverting op-amp topology, see Fig. 24.26).

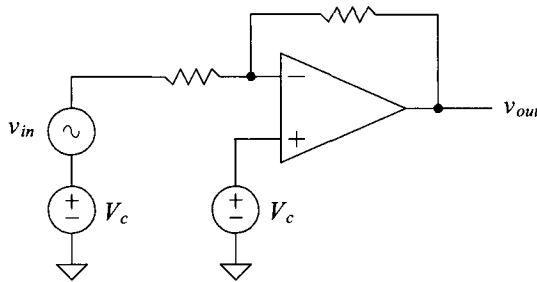


Figure 24.26 An inverting op-amp topology. The common mode voltage is held constant.

Power Supply Rejection Ratio (PSRR)

The power supply rejection ratio (PSRR) is a term used to describe how well an amplifier rejects noise or changes on the V_{DD} and or ground power buses. This parameter can be extremely important in precision analog design. Consider the test setup shown in Fig. 24.27. The positive PSRR is defined by

$$PSRR^+ = \frac{A_{OL}(f)}{v_{out}/v^+} \quad (24.28)$$

while the negative PSRR is defined by

$$PSRR^- = \frac{A_{OL}(f)}{v_{out}/v^-} \quad (24.29)$$

Ideally, v_{out} doesn't vary with changes in V_{DD} and ground (and so the $PSRR$ is infinite).

To understand how variations in V_{DD} or ground can feed to the output of the amplifier, consider the op-amp in Fig. 24.8. Variations in V_{DD} feed through to the gate of M3. Because of the circuit's symmetry, this causes the drain of M4, and thus the gate of M7, to move around. However, the source of M7 is moving the same amount, causing the v_{sg} of M7 to remain constant. Again, because of the symmetry, the output voltage moves

at the same rate. Thus, noise on V_{DD} feeds directly to the output of the op-amp. This is indicated in Fig. 24.27c, where v_{out}/v^+ is one. At higher frequencies (in the kHz range), the gate and drain of M7 get shorted together through the compensation capacitor. This, again, causes all of the noise from V_{DD} to feed directly to the output of the amplifier (see netlists at cmosedu.com). The indirect compensation scheme used in the op-amp of Fig. 24.21 can be designed for larger f_{un} and, thus, has better $PSRR$ at higher frequencies. Using the common-gate stage, Fig. 24.17, can result in an even higher $PSRR$ by isolating the compensation capacitor from the power supplies. For this reason op-amp's that use a cascode-load diff-amp, Fig. 24.29, are very useful in practical design.

Noise on ground ideally won't affect the operation of the op-amp (in either of the topologies of Fig. 24.8 or 24.21). The noise appears as a voltage variation across the current sources. In reality, at low frequencies, some of the ground noise, v^- , appears at the output (this is especially true in nanometer processes that have low r_o). However, at higher frequencies, because of the compensation capacitance connected to the output of the op-amp, ground noise contributions to the output signal decrease.

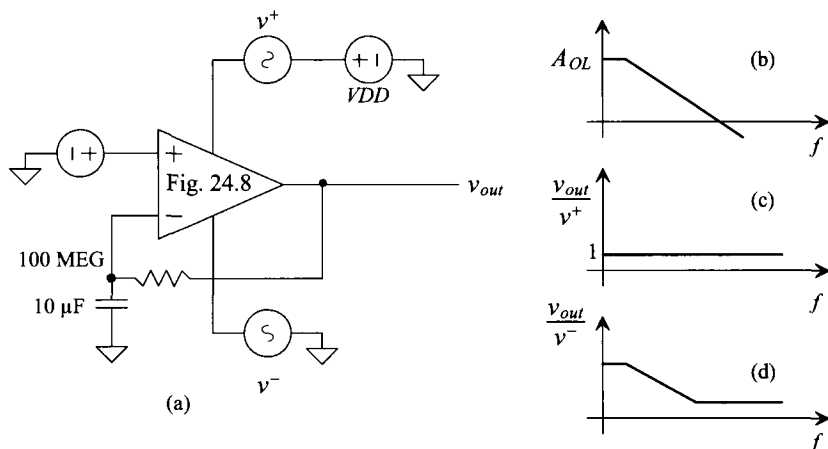


Figure 24.27 (a) Test setup to determine PSRR, (b) open-loop gain, (c) gain from AC signal on V_{DD} to output, and (d) gain from AC signal on ground to output.

Increasing the Input Common-Mode Voltage Range

It may be useful in some situations to have an input common-mode voltage range that extends close to the power supply rails. For our op-amp in Fig. 24.21, the allowable input common-mode voltage range is from (roughly) 450 mV to 900 mV (basically half of V_{DD}). Towards the goal of extending the input common-mode voltage range, consider the addition of a PMOS diff-amp to Fig. 24.21, as seen in Fig. 24.28. The circuitry on the right side of the schematic is the op-amp in Fig. 24.21 drawn a little differently. Here, we've drawn the current source of the diff-amp as two parallel current sources. In Fig. 24.21 the current source biasing the NMOS diff-amp used MOSFETs with 100/2 sizes. Here we supply the same current but use two 50/2-sized current sources. Also, we've bumped up the size of the PMOS devices (M3, M4, and M7) by two. This is to accommodate the extra current supplied by the added PMOS diff-amp (the left side of the schematic). We could keep the PMOS devices the same size as seen in Fig. 24.21 and the

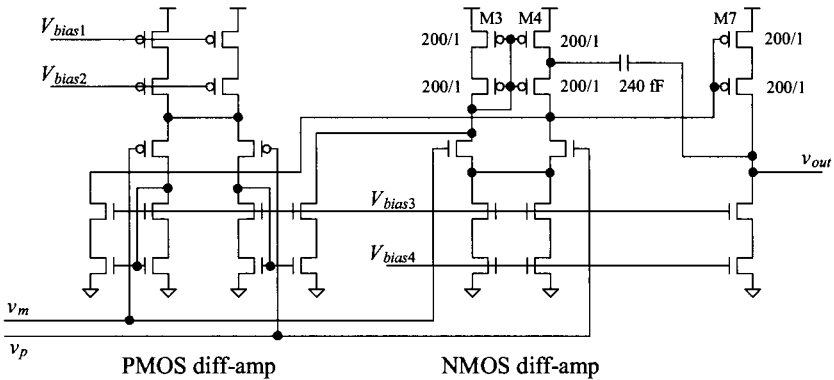


Figure 24.28 Two-stage op-amp of Fig. 24.21 with rail-to-rail input range.

circuit would still work fine (the V_{SG} voltages of the PMOS would simply be a little larger when both diff-amps are on). Two other notes: notice that the DC currents flowing in M3, M4, and M7 change depending on the input common-mode voltage. When the input common-mode voltage is large, the PMOS diff-pair is off; when it's small, the NMOS diff-pair is off. In the middle, both are conducting current. This means that the transconductance of the diff-amp will vary from g_{mn} to $g_{mn} + g_{mp}$ to g_{mp} . When both diff-amps are on (assuming $g_{mn} = g_{mp}$), the transconductance is twice as high as when a single diff-amp is on. This may require doubling the compensation capacitor. Finally, the change in the DC biasing conditions result in an offset that is a function of the input common-mode voltage.

Estimating Bandwidth in Op-Amp Circuits

When we finish designing an op-amp, its frequency response can be written using a single dominant-pole response (see Fig. 21.26 on page 680 with $A_{DC} = A_{OLDC}$) as

$$A_{OL}(f) = \frac{A_{OLDC}}{1 + j \cdot \frac{f}{f_{3dB}}} \quad (24.30)$$

For the op-amp response in Fig. 24.19, we can write

$$A_{OL}(f) = \frac{1,000}{1 + j \frac{f}{10 \text{ kHz}}} \quad (24.31)$$

To estimate the bandwidth of a closed-loop op-amp circuit where $f \gg f_{3dB}$, we can write Eq. (24.30) as

$$|A_{OL}(f)| \approx \frac{A_{OLDC}}{\frac{f}{f_{3dB}}} = \frac{A_{OLDC} \cdot f_{3dB}}{f} = \frac{f_{un}}{f} \quad (24.32)$$

noting the unity-gain frequency is calculated using

$$f_{un} = A_{OLDC} \cdot f_{3dB} \quad (24.33)$$

From this equation, we should see why the unity-gain frequency, f_{un} , is called the *gain-bandwidth product*. Note that for every increase in frequency by 10 (decade) above f_{3dB} , we get a decrease in A_{OL} by 10 (–20 dB). Alternatively, for every increase in frequency by 2 (octave) above f_{3dB} , we get a decrease in A_{OL} by 2 (–6 dB).

The closed-loop bandwidth (which we'll call f_{3dBCL}) of the op-amp circuit cannot be larger than the bandwidth of the op-amp. Therefore, to estimate the bandwidth of a closed-loop op-amp circuit, assuming the op-amp is the limiting factor, we can write

$$A_{CL} \cdot f_{3dBCL} = f_{un} = \text{gain-bandwidth product} \quad (24.34)$$

If the op-amp circuit has a closed-loop gain of one (a follower configuration as seen in Fig. (24.7)), the bandwidth is f_{un} . The op-amp in Fig. 24.21 used in a follower configuration would have a bandwidth of (roughly) 100 MHz. If this op-amp were used in a closed-loop configuration with a gain of 10, then the bandwidth of the op-amp circuit would be estimated as 10 MHz.

24.2 An Op-Amp with Output Buffer

In the previous section, the low-frequency open loop gain of the op-amps in Figs. 24.8 or 24.21 was determined by a product of the diff-amp's gain, A_1 , and the common-source's gain, A_2 or

$$A_{OLDC} = A_1 \cdot A_2 = \overbrace{g_{mn}(r_{o2} || r_{o4})}^{A_1} \cdot \overbrace{g_{mp}r_{o7}}^{A_2} \quad (24.35)$$

which, from the simulations (see Fig. 24.19 for example) was 1,000. The small-signal resistance on the output of the op-amp, r_{o7} , is (from Table 9.2) 333 k Ω . If we were to connect a 10 k Ω resistor to the op-amp output, then A_{OLDC} would drop to 33. As seen in Fig. 24.1, we can add a buffer to the output of second stage to isolate it from a load resistance (or a large capacitance). We could use a source-follower (that has a voltage gain of 1), as seen in Figs. 21.46 or 21.48. However, this addition limits the output swing of the op-amp (which may be OK in some situations). Next we could try using a push-pull topology, as seen in Fig. 21.49. However, the voltage gain of the push-pull amplifier is >1 (see Eq. (21.116)) so if we were to include it in our basic op-amp of Fig. 24.21, we would have a three-stage op-amp (meaning all three stages have voltage gains greater than one). A three-stage op-amp can be challenging to compensate over production corners (and temperature). Also, as seen in Eq. (21.116), a small load resistor (k Ω) would still kill the push-pull amplifier's gain. To keep the gain high, let's try to use a first-stage topology with a large gain (increase A_1) while using a push-pull amplifier for the second stage. The gain of the second stage can still drop but, since the gain of the first stage is large, we still end up with a reasonable overall gain.

Towards the goal of increasing the first-stage gain, consider the op-amp schematic seen in Fig. 24.29. For the first stage, we've used the cascode diff-amp originally seen in Fig. 22.30. The gain of this diff-amp (now A_1) is 500 (from Tables 9.2, and 20.1 and Eq. (22.48)). The second stage of the op-amp is the topology seen in Fig. 21.50. Its gain depends on the load resistance it drives. If no load is connected to its output, then

$$A_2 = -10 \cdot (g_{mn} + g_{mp}) \cdot \frac{(r_{op} || r_{on})}{10} = -31.6 \text{ V/V} \quad (24.36)$$

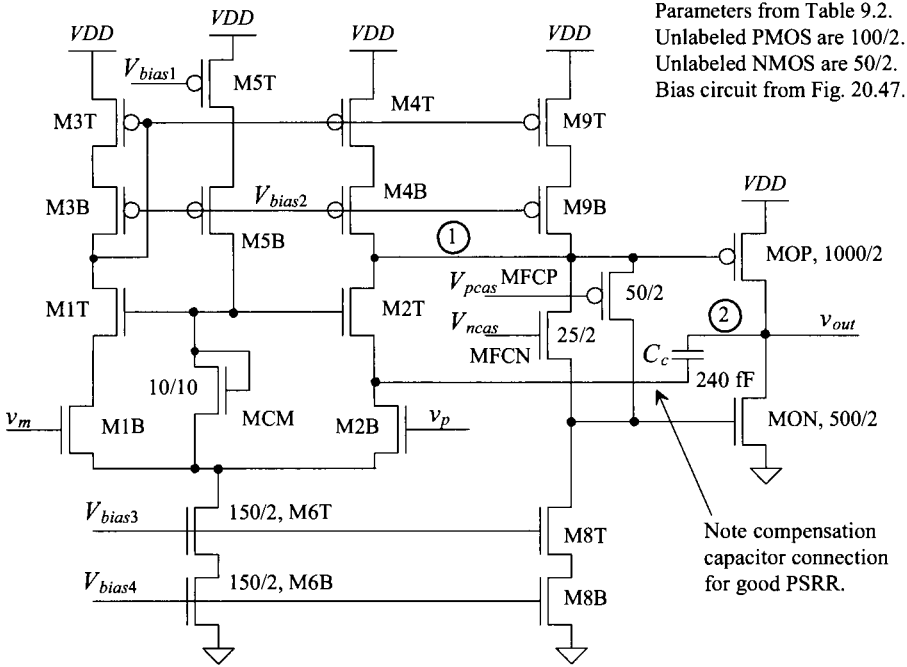


Figure 24.29 A CMOS op-amp with output buffer.

giving an overall low-frequency gain, A_{OLDC} , of 15,800 ($= 84$ dB). Note that the widths of MON and MOP are ten times wider than the values specified in Table 9.2 and so, as seen in Fig. 21.50, they conduct $100\text{ }\mu\text{A}$ of current. In Eq. (24.36) we multiplied the g_m s in Table 9.2 by 10 ($g_{mon} = 10 \cdot \beta_n (V_{GS} - V_{THN}) = 10 \cdot g_{mn, \text{Table 9.2}}$) and divided the output resistances given in Table 9.2 by 10 ($r_{omon} = \frac{1}{\lambda \cdot 10 \cdot 10\text{ }\mu\text{A}} = \frac{r_{on}}{10}$).

Compensating the Op-Amp

We increased the gain of the first stage by increasing R_1 (the resistance at node 1 is now set by cascoded current sources). As seen in Eq. (21.65), this pushes the pole lower in frequency. As just described in Eq. (24.33), this decrease in f_1 ($= f_{3dB}$) and increase in low-frequency gain *doesn't have any effect* on the gain-bandwidth product ($= f_{un}$) of the op-amp. What this indicates, to a certain extent, is that we can still compensate the op-amp with a 240 fF capacitor as we did in the last section. To feed the indirect compensation current to node 1 we connect the compensation capacitor to the source of M2T, as seen in Fig. 24.29. This ensures good PSRR.

The open-loop response of the op-amp in Fig. 24.29 (without a load) is seen in Fig. 24.30. The phase-margin is roughly 70 degrees. The step response of the op-amp driving a $1\text{ k}\Omega$ resistor and a 10 pF capacitor is seen in Fig. 24.31. The input signal is a pulse from 100 mV to 900 mV . Since the gain of the amplifier is -1 and the common-mode voltage is 500 mV , the output swings from 900 mV to 100 mV .

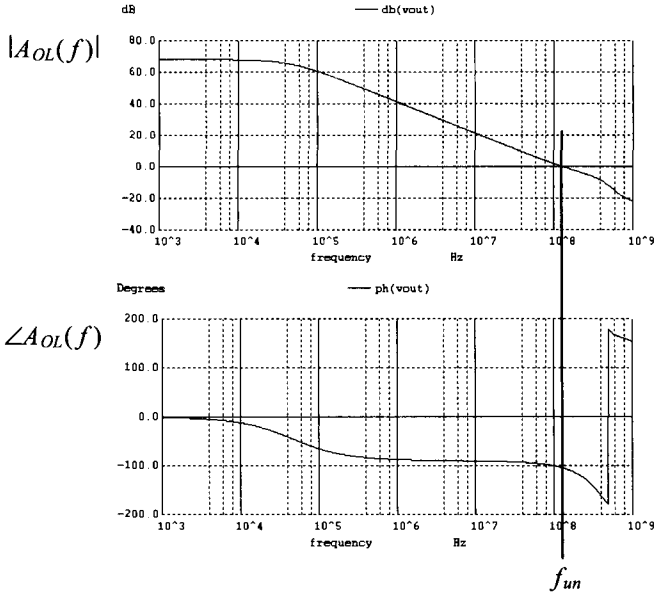


Figure 24.30 Open-loop response of the op-amp in Fig. 24.29.

This is a good point to remember one of the fundamentals we presented in Ch. 9: that is, for high-speed design, we must use minimum length devices (see Eq. (9.59)). If we reduce all of the length of 2 (100 nm) devices in Fig. 24.29 to length of 1 (50 nm) devices, we can increase the speed of the op-amp. Figure 24.32 shows how the step response in Fig. 24.31 changes when we make this reduction in length. The big problem with using minimum channel lengths is the reduction in gain. Close inspection of Fig. 24.32 shows that the output is never quite pulled up to the correct voltage (it is either slightly below 900 mV or 100 mV). Simulating the AC open-loop response results in a gain of 44 dB (= 159). An output voltage of 900 mV would result in a difference on the inputs of the op-amp of 5.66 mV ($0.9/159 = v_p - v_m$).

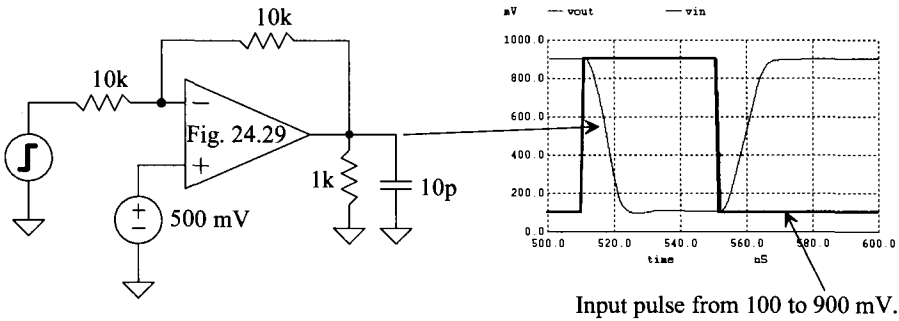


Figure 24.31 Step response of the op-amp in Fig. 24.29 driving 1k and 10 pF.